

MADHAV DEEPAK

B.TECH COMPUTER SCIENCE STUDENT | SOFTWARE ENGINEERING & APPLIED AI

Amaravati, Andhra Pradesh | maadhavdeepak@gmail.com | 8714167468

Portfolio madhavdeepakk.github.io

GitHub github.com/madhavdeepakk

LinkedIn linkedin.com/in/madhav-deepak-3190a13b1

SUMMARY

Final-year B.Tech Computer Science student who builds end-to-end software: from data ingestion and REST APIs to ML/LLM components and deployed web front ends. Currently building an AI system for supply-chain and procurement teams that detects commodity price anomalies and explains them with retrieval-grounded, cited evidence. Strong in data structures, algorithms and Python, with hands-on work in FastAPI, JavaScript, graph and vector databases, and cloud deployment. Owns work from design to deployment and validates results against baselines.

EDUCATION

Qualification	Institute Name	University / Board	Year of Passing	Score	Course Type
B.Tech – Computer Science	VIT-AP University, Amaravati	VIT-AP University	2027 (expected)	CGPA 7.95 / 10 (till 6th sem)	Full-time
Class XII (HSC)	GEMS Millennium School, Sharjah	CBSE	2023	83%	Full-time
Class X (SSC)	Our Own English High School, Sharjah – Boys Branch	CBSE	2021	89.6%	Full-time

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML, CSS

Frameworks, Libraries & Databases: FastAPI, Node.js, React, Streamlit, Hugging Face, ONNX Runtime, OpenCV, PostgreSQL, Redis, FAISS, Neo4j

Applied AI / ML: Retrieval-Augmented Generation (RAG), Time-Series Anomaly Detection, MICE Imputation, Class Imbalance, Cross-Validation

Cloud, Tools & Platforms: Docker, Kubernetes, GitHub Actions, Git, Vercel, Neo4j Aura, MQTT

CERTIFICATIONS

Artificial Intelligence Fundamentals – IBM • AWS Academy Cloud Foundations – AWS • AWS Academy Cloud Architecting – AWS

CORE SUBJECT PROFICIENCIES

- Data Structures & Algorithms
- Object-Oriented Programming (Java, Python)
- Machine Learning
- Database Systems (Graph & Vector)
- Software Engineering & REST API Design
- IoT Systems (ESP32, MQTT)

STRENGTHS

- Takes end-to-end ownership, from scoping to shipped product
- Execution-focused: plans in milestones, builds baselines first
- Highly collaborative team player with strong interpersonal skills
- Iterative, experimentation-driven approach to problem-solving
- Clear communicator of technical concepts in team settings
- Quick to learn new technologies and tools

LANGUAGES

- English – Full Professional Proficiency
- Malayalam – Full Professional Proficiency
- Hindi – Limited Working Proficiency

PROJECTS UNDERTAKEN

Commodity Disruption Signal & Explanation System | Python, Hugging Face, FAISS, Streamlit

Final-year capstone · team of 4 · in progress (2026)

- Building an AI-driven risk-monitoring pipeline for supply-chain and procurement teams that flags significant arabica coffee price anomalies and explains them using cited public evidence
- Integrating 5 public data sources (NOAA weather, GDELT news, Freightos shipping, World Bank, FRED macro) with daily-to-monthly update schedules into one timezone-normalized dataset, stamping every value with its release date to prevent lookahead bias
- Flagging anomalies with a rolling-volatility z-score (60–90 day window) instead of a fixed % threshold; forecast model must beat a naive baseline under walk-forward validation
- Designing a RAG explanation layer on a FAISS vector index with relevance gating: the LLM must cite retrieved evidence or return “insufficient evidence”, cutting hallucinated explanations
- Shipping a Streamlit dashboard and a daily scheduled pipeline (GitHub Actions) with logging; measuring precision, recall and false-positive rate on historical disruptions and no-event control periods

CrowdWatch — Real-Time Overcrowding Detection | Python, YOLOX, ONNX Runtime, FastAPI, Docker

- Built a CPU-only system that counts people per zone from CCTV video and warns before a zone fills up
- Combined a YOLOX detector, a custom ByteTrack-style tracker and a DM-Count density model, cutting counting error on the ShanghaiTech Part B benchmark from 100 to 13 people per image
- Caught 47 of 51 overcrowding events with 47 alarms in 60 simulated runs (a per-frame threshold raised 2,920); shipped with a live demo, CI and 89 automated tests

Smart Resume Screener | Python, FastAPI, JavaScript

- Engineered an automated full-stack platform with a FastAPI backend and Vanilla JS/HTML5/CSS3 front end to evaluate and rank candidate profiles against job specifications in real time
- Turned unstructured resume text into a structured comparison model, exposed to the front end through API endpoints

Full-Stack Todo App | Node.js, React, PostgreSQL, Redis, Docker, Kubernetes

- Built a task manager with an Express REST API on PostgreSQL and a React front end, with JWT and Google OAuth sign-in, Redis caching and Razorpay payment integration
- Containerized with Docker Compose and wrote Kubernetes manifests with API autoscaling

ML Pipeline for Imbalanced Classification | Python

- Built an end-to-end ML pipeline to predict exoplanet habitability from a severely imbalanced dataset
- Diagnosed and removed a target-leakage feature that was artificially inflating model performance, catching a data quality issue before it reached production
- Handled high missingness and class imbalance using MICE imputation and a class-weighted Random Forest; applied stratified cross-validation and threshold tuning to improve minority-class recall over baseline